

How to Fulfill This Paper: "Imbalance-Aware Machine Learning for Anomaly Detection in Nuclear Power Plant CPS Data"

This is a complete execution roadmap — from where you stand today (dataset explored, six-model pipeline built, correlation finding known) to a submission-ready manuscript. It's organized as sequential phases you can work through in order.

Phase 0: Lock the Paper's Core Claim (do this before writing any more code)

Before touching code further, write a **one-paragraph abstract draft** — even a rough one. This forces you to commit to what the paper actually argues, which should be:

"We systematically evaluate six ML models under four imbalance-correction strategies on a Nuclear Power Plant CPS anomaly detection dataset. Despite comprehensive correction and tuning, performance remains near chance level across all configurations. We show through correlation, mutual information, and SHAP-stability analysis (including a label-permutation control) that this ceiling is not an imbalance problem but a label-feature independence problem — with implications for how synthetic CPS benchmarks should be validated before use."

Everything below exists to produce the evidence for that paragraph. Keep it pinned somewhere visible — it's your filter for "does this experiment belong in the paper."

Phase 1: Finish the Experimental Pipeline (~1-2 weeks of work)

You already have: 6 models, SMOTE, stratified 70/15/15 split, nine-phase structure. To fulfill this specific title, extend it as follows:

1. **Add the 3 missing imbalance regimes** (no-correction baseline, class-weighting, ADASYN) alongside your existing SMOTE runs — same 6 models $\times$ now 4 regimes = 24 runs minimum.
2. **Swap primary metric emphasis** — make PR-AUC and MCC the headline numbers in every table; keep ROC-AUC but label it secondary (state why: uninformative at ~5% prevalence).
3. **Add statistical testing** — DeLong's test (ROC-AUC), McNemar's test (paired predictions), bootstrap 95% CIs (2,000 resamples) on every metric. This is what separates a rigorous paper from a script that prints numbers.
4. **Run the diagnostic layer** — you've already done correlation; add mutual information per feature, and a Bayes-error / k-NN entropy estimate if you want a theoretical ceiling number (optional but strong).
5. **Run SHAP + the shuffled-label control** — this is the evidence for your central claim. Compute SHAP on real labels and on 10-20 shuffled-label reruns; compare top-k

feature stability (Jaccard overlap) between the two. If they're statistically indistinguishable, that's your proof.

6. **Log everything systematically** — one results table per (model $\times$ regime), one master CSV with every run's metrics + CIs. You'll draw directly from this for the Results section.

Phase 2: Turn Results Into the Paper's Evidence Base

Once Phase 1 runs are done, produce these four artifacts — they become your core figures/tables:

Artifact	What it shows	Where it goes
Master performance table	All 6 models $\times$ 4 regimes $\times$ 5 metrics + CIs	Results §1
Significance matrix	Pairwise DeLong's/McNemar's p-values across model/regime pairs	Results §2
Correlation + MI table	Per-feature correlation and mutual information with the label	Results §3 (data audit)
SHAP stability comparison	Real-label vs. shuffled-label attribution stability (bar chart or heatmap)	Results §4 (core novel result)

If any regime/model combination *does* beat chance meaningfully (check this honestly — don't assume the answer before running Phase 1 in full), that becomes your most important result and the abstract should be adjusted to feature it.

Phase 3: Write the Manuscript

Standard IMRaD structure, sized for an IEEE conference or Scopus Q2 journal (6–10 pages):

1. **Introduction (~1 page)** — NPP CPS security motivation, why imbalance matters in this domain, the specific gap: imbalance-correction comparisons are largely absent from NPP-CPS literature, and none of it audits label-feature dependence before reporting results.
2. **Related Work (~1 page)** — your 10-paper table, condensed to prose + the table itself.
3. **Dataset (~1 page)** — full description, class distribution, the audit subsection (correlation/MI) presented as methodology, not a caveat.
4. **Methodology (~1.5 pages)** — preprocessing, the 4 imbalance regimes, 6 models, tuning/CV protocol, metrics, SHAP + permutation-control procedure.

5. **Results (~2 pages)** — the four artifacts from Phase 2, described plainly with numbers, not adjectives.
6. **Discussion (~1 page)** — this is where you make the argument: performance doesn't improve with better imbalance correction because the ceiling is set by label-feature independence, not class skew. Discuss implications for anyone using this dataset or similar synthetic CPS benchmarks.
7. **Conclusion + Future Work (~0.5 page)** — point to physics-informed features, real/simulator-grounded data, and the audit framework as a reusable pre-modeling step.

Writing order that works well in practice: Methodology → Results → Discussion → Introduction → Related Work → Conclusion → Abstract last. Writing the abstract last (even though you drafted a rough one in Phase 0) ensures it accurately reflects what you actually found.

Phase 4: Internal Review Before Submission

- **Send to your supervisor** at the full-draft stage, not before — you already have their stratified-split requirement satisfied, so this is the natural checkpoint.
- **Self-check against the pre-registered outcome definitions** (from the earlier XAI methodology) — make sure the Results/Discussion don't quietly reinterpret what "stable" or "meaningful" means after seeing the numbers.
- **Check every number in the abstract/conclusion appears verbatim in a table** — a common reviewer complaint is unsupported summary claims.

Phase 5: Venue Selection & Submission

Given the paper's actual contribution (rigorous benchmark + diagnostic finding, not a novel algorithm):

Venue type	Fit	Notes
IEEE conference (e.g., regional/energy/CPS-security tracks)	Best fit	Conferences readily accept well-executed benchmark + diagnostic studies
Scopus Q2 journal (e.g., <i>IEEE Access</i> , <i>Applied Sciences</i>)	Good fit	Look for special issues on CPS security, data-centric AI, or industrial anomaly detection
Scopus Q1 / top nuclear-engineering journals	Unlikely without real/simulator data	Keep as a future-work target once real data is available

Before submitting: check the target venue's current call for papers and formatting template — I can search for open CFPs matching your topic and timeline if you tell me a rough deadline window.

Suggested Timeline

Week	Task
1-2	Phase 1: complete the 4-regime $\times$ 6-model runs, statistical tests, SHAP + shuffled-label control
3	Phase 2: build the four core artifacts (tables/figures)
4	Phase 3: full manuscript draft
5	Phase 4: supervisor review + revisions
6	Phase 5: venue selection, formatting, submission

What I Can Help With Next

- Draft the actual notebook cells for the 4-regime $\times$ 6-model grid (extends your existing pipeline directly)
- Write the SHAP + shuffled-label-control code
- Search for currently open CFPs (IEEE/Scopus Q2) matching your topic and timeline
- Draft any individual section of the manuscript once Phase 1 results exist